

Tanvir Ahmed

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RESEARCH FOCUS

Mechanical & Manufacturing Engineering Ph.D. researcher specializing in wearable, multi-sensor systems and applied machine learning on time-series sensor data, with six publications. Designed, built, and field-deployed a custom-PCB smart collar (6-axis IMU, GNSS, BLE/Wi-Fi) for cattle behavior classification, collecting and video-labeling a multi-day sensor dataset, benchmarking classical and deep-learning models (Random Forest, XGBoost, CNN-BiLSTM), and deploying the best model to a real-time FastAPI dashboard demonstrating end-to-end capability from sensor hardware to deployed ML. Additional experience in human biomechanical-signal analysis from body-worn IMUs (gait characterization) and deep-learning perception for field-robot navigation.

EDUCATION

Ph.D., Mechanical & Manufacturing Engineering Aug 2025 – May 2029 (Expected)

Texas State University, San Marcos, TX | GPA: 4.00 / 4.00 (through Spring 2026)

- Relevant Coursework: Advanced Robotics (CS 7389K), ML/AI for Engineers (MMIE 7310), Applied Data Science 1 (MMIE 7317), Advanced Design of Experiments (MMIE 7305).

M.S., Mechanical & Manufacturing Engineering Aug 2023 – Aug 2025

Texas State University, San Marcos, TX | GPA: 4.00 / 4.00 | Automation and Robotics Lab

- Thesis Title: Development of a wearable sensor unit for cattle behavior prediction using machine learning.
- Thesis Supervisor: Dr. Heping Chen
- Relevant Coursework: Computing for Data Analytics (CIS 5357), Advanced Robotics (MFGE 5326), Advanced CAD/CAM (MFGE 5316), Machine Learning for Engineering Applications (EE 5331), Probability, Random Variables, & Stochastic Processes for Engineers (ENGR 5310)

B.S., Mechanical Engineering Jan 2015 – Nov 2018

Islamic University of Technology, Dhaka, Bangladesh | GPA: 3.39 / 4.00

RESEARCH INTERESTS

Wearable & Multi-Sensor Systems, Time-Series Machine Learning, Human-Robot Interaction, Applied ML & Perception, Robotics & Automation, Precision Agriculture & Agricultural Robotics

RESEARCH EXPERIENCE

ML-Based Wearable Sensor Unit for Cattle Behavior Prediction TXST · USDA-NIFA

- Designed and built a custom 2-layer KiCad PCB smart collar (6-axis IMU, GNSS, BLE/Wi-Fi) in a waterproof enclosure (~31.5 h active battery life); field-deployed at Freeman Center, Texas under IACUC protocols.
- Collected and video-labeled a multi-day sensor dataset (~356k samples, 4 behaviors); benchmarked Random Forest, XGBoost, and CNN-BiLSTM (SMOTE + FFT features) and deployed the best model (RF, 85.4% accuracy) to a real-time FastAPI dashboard.

IMU-Based Human Gait Analysis of Walking and Running TXST · USDA-NIFA

- Analyzed walking and running gait using two foot-mounted IMUs (WitMotion WT901): applied Butterworth low-pass filtering, segmented gait cycles via peak detection, and extracted time-domain statistical features to characterize biomechanical differences for human health-monitoring applications (IEEE CYBER 2024).

Deep-Learning Crop-Row Perception for Field-Robot Navigation TXST

- Built a crop-row segmentation pipeline (U-Net with a ResNet-34 encoder, 0.820 IoU) with geometric heuristic refinement to generate navigation guidance for autonomous field robots.

Passive Self-Righting Robot Design for Cattle Farming (AnimalCareBot) TXST · USDA-NIFA

- Developed motor control and electrical-system setup for the AnimalCareBot self-righting robot, and fabricated mechanism components (lathe, CNC) as part of a team project.

Additional projects: TPMS lattice fabrication via multi-axis CNC machining; magnetic characterization of PA-4.6 bonded-magnet filaments developed using twin screw extrusion (NSF); blockchain-based Sybil-attack detection in underwater sensor networks; activated carbon from agricultural waste.

PUBLICATIONS (6 publications; 46 citations, Google Scholar)

1. M. M. Alam, M. A. Kabir, M. F. A. Omei, S. Ahmed, **T. Ahmed**, M. Y. Arafat, "Integrated Trajectory and Resource Allocation for Air-Ground Networks," *IEEE Open Journal of the Communications Society*, 2026.
2. M. S. U. Khan, **T. Ahmed**, H. Chen, "A Review of SLAM Techniques for Agricultural Robotics: Challenges, Architectures, and Future Directions," *IEEE CYBER*, 2025.
3. **T. Ahmed**, M. S. U. Khan, H. Chen, "A Review of IoT for Animal Health Farming," *IEEE CYBER*, 2025.
4. P. Karkhanis, O. K. Arigbabowo, A. J. Gonzalez, S. Inamdar, **T. Ahmed**, A. Herrera, W. J. Geerts, J. S. Tate, "Magnetic Properties Evaluation of Polyamide 4.6 Bonded Magnetic Composite," *SAMPE Conference*, 2024.
5. **T. Ahmed**, H. Chen, P. Panta, "Human Motion Gait Analysis Using IMU Sensors," *IEEE CYBER*, 2024.
6. M. M. Arifeen, A. Al Mamun, **T. Ahmed**, et al., "A Blockchain-Based Scheme for Sybil Attack Detection in Underwater Wireless Sensor Networks," *Springer AISC*, 2021.

POSTERS & PRESENTATIONS

1. "Autonomous Navigation and Control of an Animal Care Robot Using SLAM," TXST Graduate Student Research Conference (GSRC) 2024 and TXST STEM Conference 2025.
2. "Development of a Self-Righting Mobile Robot and Wearable Sensor Unit for Cattle Monitoring and Behavior Analysis," U.S. DOE EM Symposium (Honorable Mention) and TXST CoSE Advisory Council Meeting, 2025.
3. "Magnetic Properties Evaluation of Polyamide 4.6 Bonded Magnetic Composite," TXST GSRC 2024.

TEACHING & PROFESSIONAL EXPERIENCE

Doctoral / Graduate Instructional & Research Assistant

Aug 2023 – Present

Ingram School of Engineering, Texas State University

- Serve as TA for robotics, automation, and controls courses: Intelligent Robotics and Control (MFGE 4176), Introduction to Industrial Robotics (MFGE 4378), Advanced Robotics in Manufacturing Automation (MFGE 5326), Control Systems and Instrumentation (MFGE 4376), and Tool Design (MFGE 4365).
- Assisted students in labs covering PID control, FANUC and ABB robotic arms, and PLC programming.
- Contributed to three research projects under Dr. Heping Chen, resulting in multiple publications.
- Graded assignments, exams, and projects for 200+ students across 7 undergraduate/graduate courses and labs.

Co-Founder & Project Manager, Monochrome Limited | Bangladesh

Jul 2021 – Aug 2023

- Led a 5-member team across project execution, financial management, and client delivery for a digital marketing firm.

TECHNICAL SKILLS

Machine Learning & AI: Python (NumPy, pandas, scipy.signal, scikit-learn, TensorFlow, PyTorch); deep learning (U-Net, ResNet, CNN-BiLSTM); image segmentation; multimodal sensor-data and time-series classification; signal processing (Butterworth filtering, FFT, peak detection, time-domain feature extraction).

Robotics & Control: Embedded sensing & multi-sensor systems (custom PCB design in KiCad, IMU/GNSS, BLE/Wi-Fi telemetry); microcontroller firmware (Arduino/C++); real-time data streaming (BLE/Wi-Fi/UDP/HTTP); ML model deployment & dashboards (FastAPI, Streamlit, Plotly). *Instructional (TA-level):* PID control, PLC programming, FANUC & ABB robotic arms. *Familiar with:* ROS2, Gazebo (TurtleBot).

Computing & Tools: Linux/Bash, High-Performance Computing; MATLAB; LabVIEW; SolidWorks; Fusion 360; KiCad (PCB); Mastercam. Fabrication: CNC, lathe, milling, 3D printing, waterjet.

Research Methods: Experiment design and study protocols (IACUC, field data collection); statistical analysis (hypothesis testing, ANOVA) in Python; data annotation and video-based labeling pipelines.

AWARDS, HONORS & ACTIVITIES

- Graduate Merit Fellowship, Texas State University. (Fall 2025, Spring 2026)
- Featured Research Profile, "[Intelligent Agriculture](#)," TXST Hillviews Magazine (2026).
- Phi Kappa Phi and Alpha Chi National Honor Societies (top 10% academic performance).
- Sole departmental nominee, TXST CoSE Advisory Board Meeting (2025).
- Full Tuition Waiver and OIC Scholarship at the Islamic University of Technology (IUT), Bangladesh.
- Certifications & activities: Certified Supply Chain Analyst (CSCA), ISCEA; TXST CADS workshops (HPC 2024, AI/ML Education Series 2024); Member, IEEE & SAMPE; Graduate Ambassador, TXST.